

The hypothetical function of FAM131A

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FAM131A

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Abstract

FAM131A is a **cytosolic adaptor**, may **coordinate vesicle-dependent signaling in neuronal and immune systems**. Evidence from protein interactions, motif analysis, expression patterns, and genomic alterations supports its role

Introduction

☐	131408	FAM131A	NM_144635.5 MANE Select	2,437	NP_653236.3	366	protein FA	⋮
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FAM131A:

Homo sapiens family with sequence similarity 131 member A

Gene synonym: C3orf40; FLAT715; PRO1378

Length: 2437 bp

Gene ID: 131408

Gene type: protein-coding

Protein ID: NP_653236.3,

NCBI Reference Sequence (mRNA): NM_144635.5

Chromosome: 3

Map: 3q27.1

Introduction

6 exons:

- Exon1: 1-119 bp
- Exon2: 120-262bp
- Exon3: 263-356bp
- Exon4: 357-539bp
- Exon5: 540-656 bp
- Exon6: 657-2437 bp

intrinsically disordered region (IDR): 1055..1129

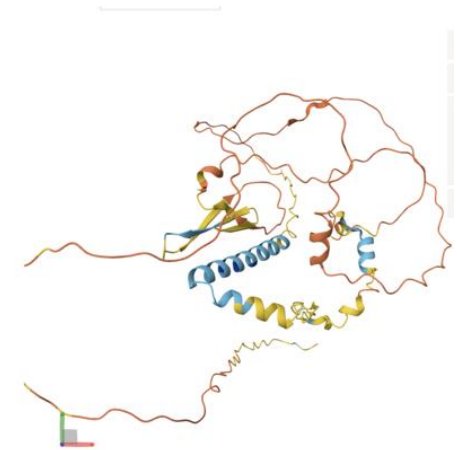
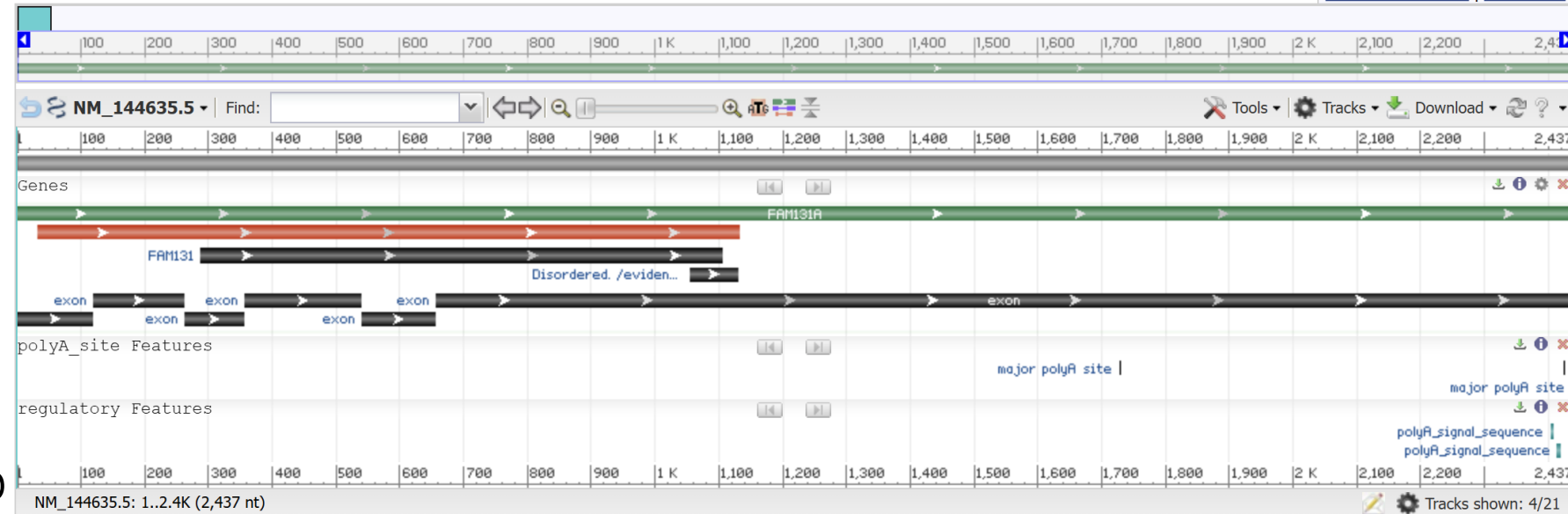
Poly A site: 1729 bp、2437 bp

Homo sapiens family with sequence similarity 131 member A (FAM131A), transcript variant 1, mRNA

NCBI Reference Sequence: NM_144635.5

[GenBank](#) [FASTA](#)

[Link To This View](#) | [Feedback](#)

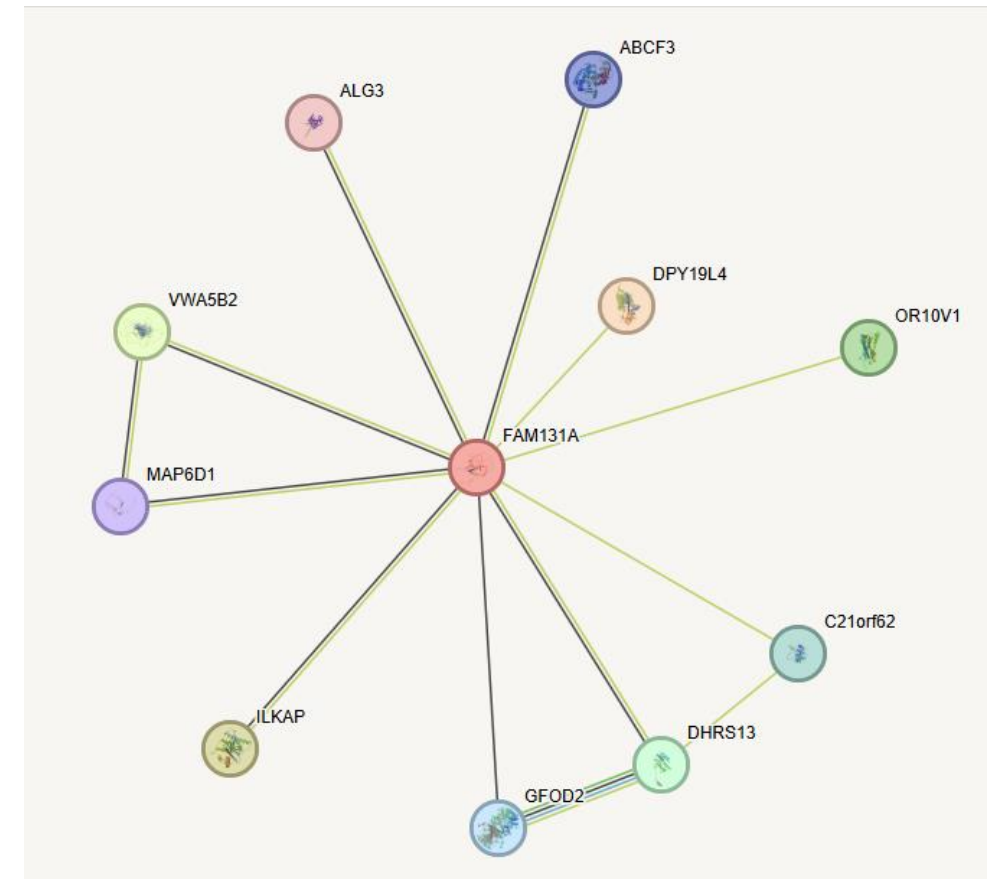


Core hypothesis

FAM131A acts as a **cytosolic adaptor** linking RNA regulation, ubiquitin signaling, and metabolism to control vesicle signaling in neurons and immune cells.

Interactor from String

Interactor	Known/Predicted Function
ALG3	Mannosyltransferase in N-linked glycosylation
ABCF3	ABC transporter family member
DPY19L4	Glycosyltransferase-like protein
OR10V1	Olfactory receptor
C21orf62	Uncharacterized protein
DHRS13	Short-chain dehydrogenase/reductase
GFOD2	Glucose–fructose oxidoreductase domain-containing protein
ILKAP	Serine/threonine phosphatase
MAP6D1	Microtubule-associated protein
VWA5B2	von Willebrand factor A domain protein

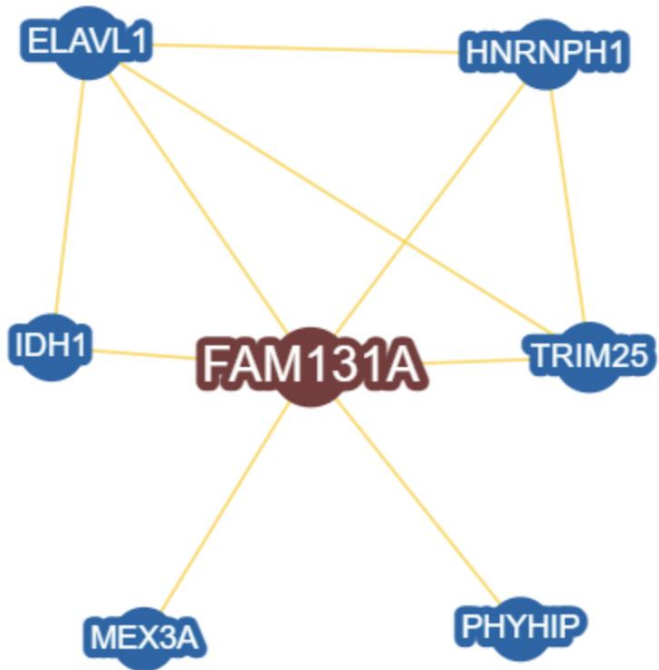


Relevance to hypothesis:

Interactors place FAM131A at the **intersection of protein processing, metabolic sensing, and cytoskeletal trafficking**, consistent with a cytosolic adaptor linking RNA regulation, ubiquitin signaling, and metabolism to control vesicle signaling in neurons and immune cells.

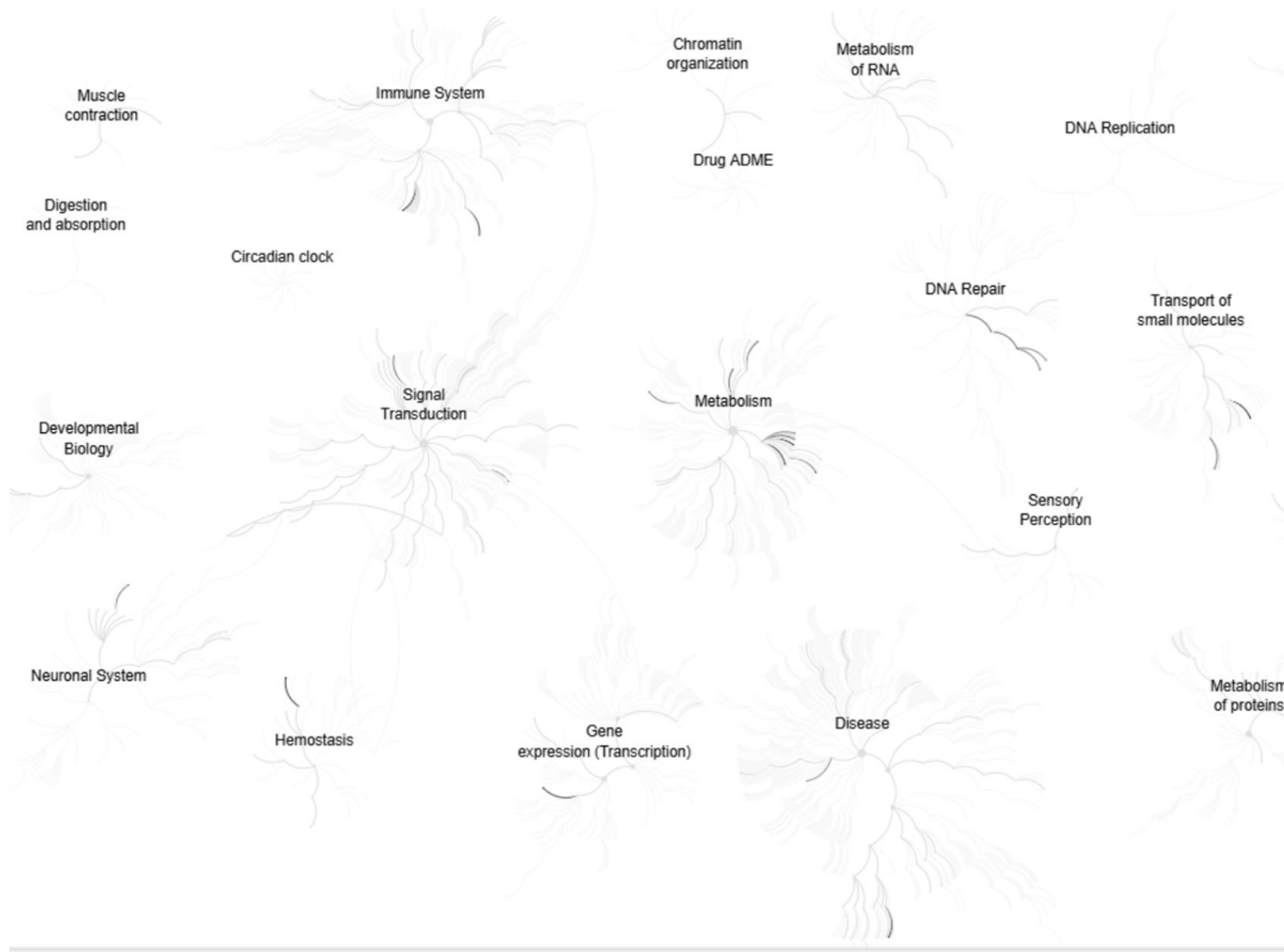
Partners in the BioGRID

Partners	Function	Relationship
ELAVL1	RNA metabolism	Direct support for the RNA–protein regulation component of the model; HuR often acts as a hub linking stress signaling to protein synthesis
HNRNPH1	RNA metabolism	Strengthens the RNA regulation angle , suggesting FAM131A associates with RNA metabolism machinery
PHYHIP	neuronal development	Introduces a neuronal context, aligning with the model's focus on synaptic vesicle maturation and exocytosis .
MEX3A	RNA metabolism	Provides dual evidence for RNA regulation and ubiquitination , a central clause in the hypothesis
TRIM25	RNA metabolism	Anchors the ubiquitin regulation theme and links FAM131A to immune receptor
IDH1	redox control.	Directly supports the redox/metabolic coupling element, consistent with the NRF2 axis mentioned in the model



BioGRID partners link FAM131A to **RNA metabolism, neuronal development** and **redox control**, supporting its role as a cytosolic adaptor connecting RNA regulation, ubiquitin signaling, and metabolism to control vesicle signaling in neurons and immune cells.

Pathway in Reactome



Pathway in Reactome

Hypothesis linkage:

FAM131A acts as **a cytosolic adaptor linking multiple major systems and pathways**—including the immune system, neuronal system, metabolism, RNA metabolism, DNA repair, small-molecule transport, signal transduction, and disease-related processes

Reactome Pathway	Functional Category
NFE2L2 regulating TCA cycle genes	Metabolic/Redox
NADPH regeneration	Metabolic/Redox
Abnormal conversion of 2-oxoglutarate to 2-hydroxyglutarate	Metabolic/Redox
Interconversion of 2-oxoglutarate and 2-hydroxyglutarate	Metabolic/Redox
Defective ALG3 causes CDG-1d	Protein Processing
ALKBH2/ALKBH3 mediated reversal of alkylation damage	DNA Repair/Epigenetics
TET1/2/3 and TDG demethylate DNA	DNA Repair/Epigenetics
NOTCH4 activation and signal transmission to the nucleus	Immune/Developmental Signaling
CLEC7A (Dectin-1) induces NFAT activation	Immune Signaling

Phosphorylation site from Eukaryotic linear motif

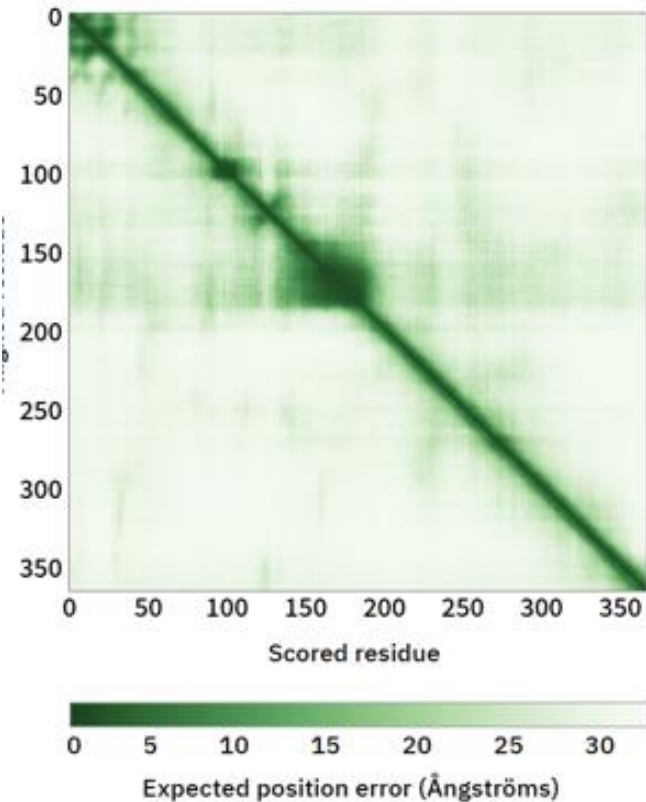
Motif (ELM ID)	Sequence (FAM131A)	Position	Mechanistic meaning
LIG_DLG_GKlike_1	RGLS LSSL	45–52	Short phospho-dependent ligand for MAGUK GK-like domains (e.g., PSD-95/DLG family) that scaffold the postsynaptic density .
MOD_CMANNOS	WKGW	135–138	C-mannosylation consensus (W..W) on Trp; a glycosylation that occurs on proteins entering/trafficking through the ER.

FAM131A contains a phospho-dependent MAGUK GK-like ligand motif. **This motif is linked to the postsynaptic density and controls vesicle maturation, docking, and receptor signaling.**

It also contains a C-mannosylation consensus sequence. **This sequence is linked to ER trafficking and affects receptor and vesicle cargo maturation.**

These motifs support its role as **a cytosolic adaptor that controls vesicle signaling in neurons and immune cells.**

Protein and mutation in AlphaFold



Predicted Aligned Error (PAE)

The PAE map shows high uncertainty for most long-range residue alignments, suggesting **a flexible, non-globular scaffold**.

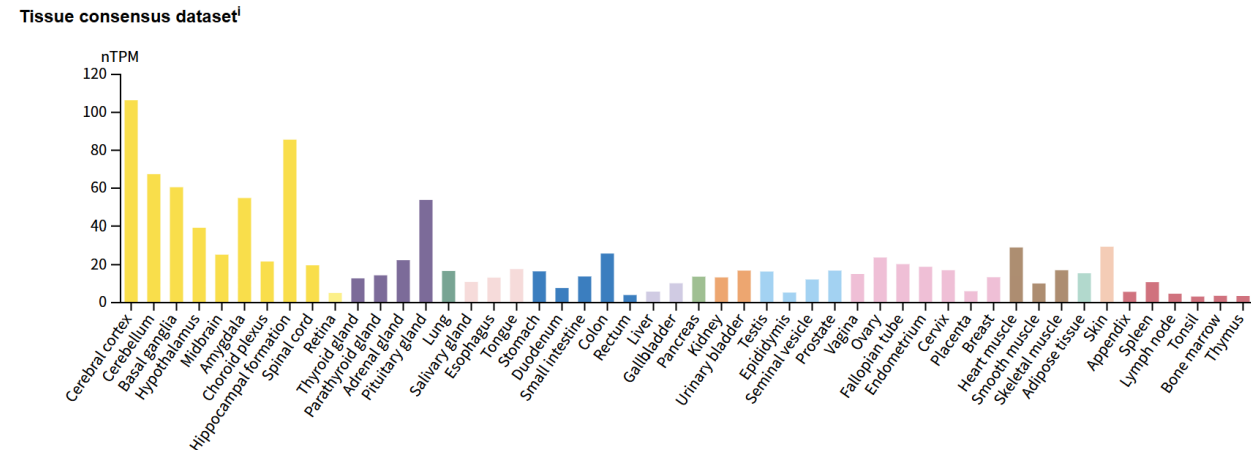
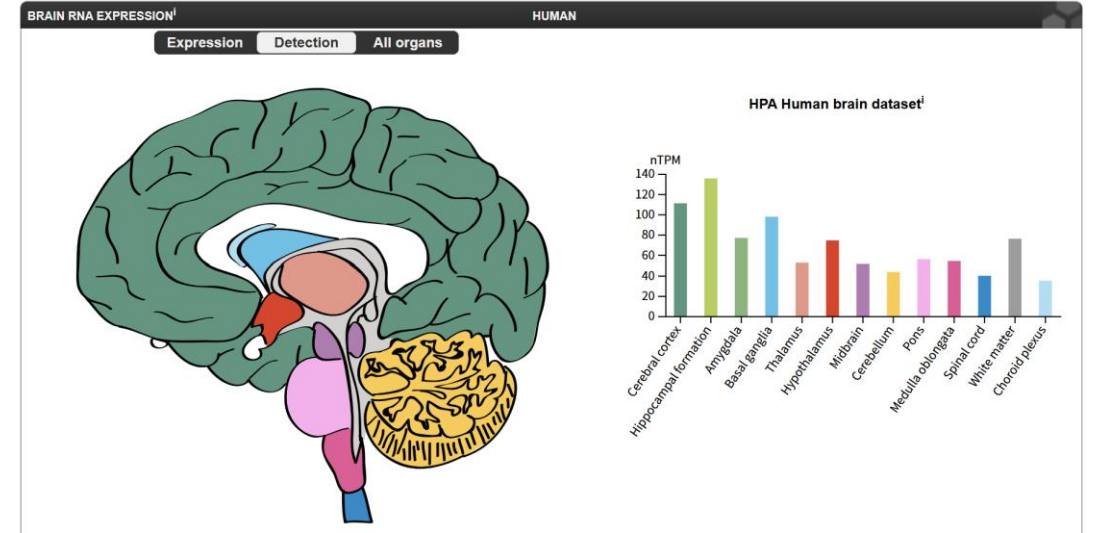
Structure identifies multiple disordered stretches **enriched in short linear motifs**

These flexible, motif-rich regions are **ideal for regulatory interactions and partner recruitment**, which supports FAM131A's role as a cytosolic adaptor controlling metabolism in order to control vesicle signaling in neurons and immune cells.



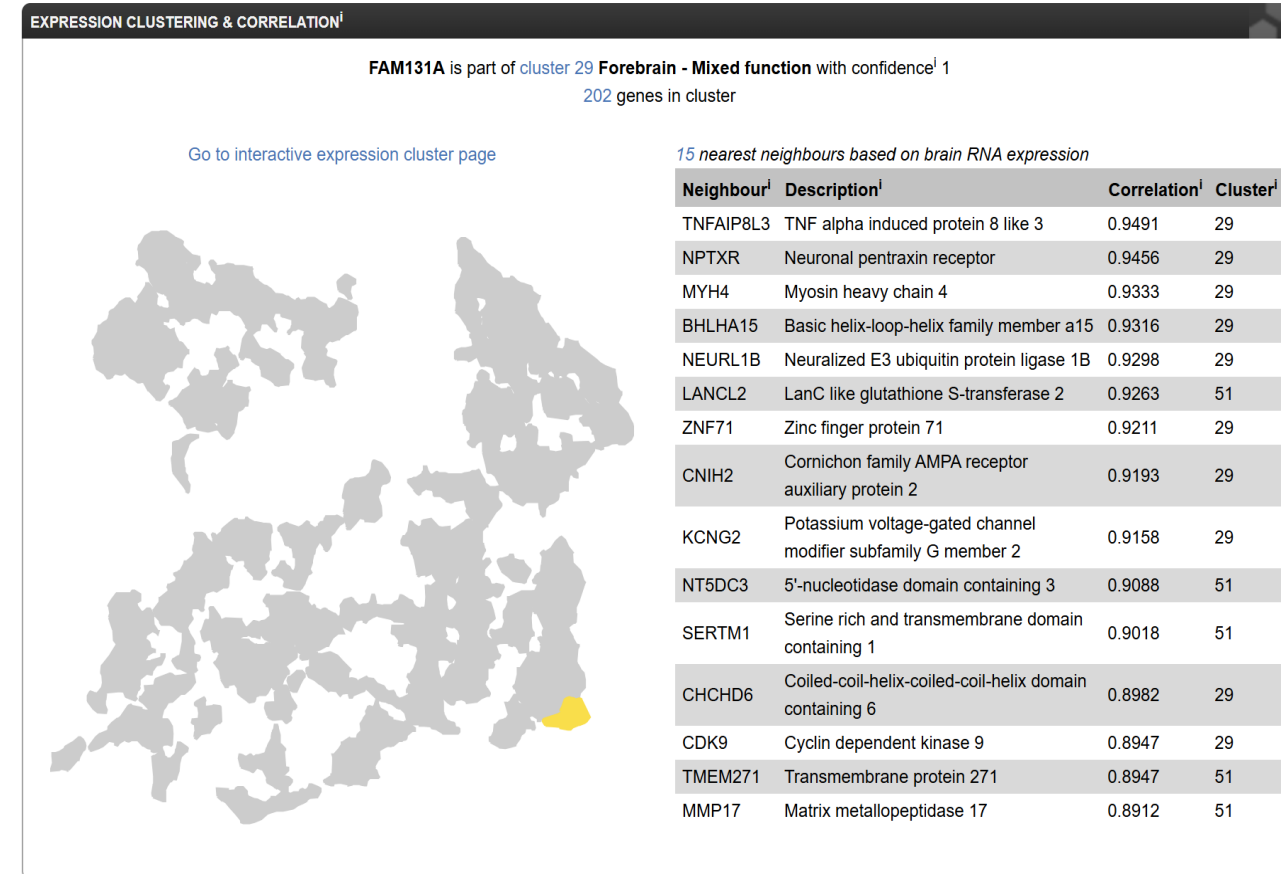
Protein in The human protein atlas

FAM131A shows **highest RNA expression in brain**, supporting a role in synaptic vesicle signaling. **Moderate expression in various peripheral tissues and detectable levels in immune organs** support its role as a cytosolic adaptor linking RNA regulation, ubiquitin signaling, and metabolism to control vesicle signaling in both neuronal and immune contexts.



Protein in The human protein atlas

FAM131A's co-expression cluster comprises genes linked to **synaptic receptor function**, **neuronal excitability**, and **ubiquitin-linked regulation**, supporting its role in integrating ubiquitin signaling, and metabolism to coordinate vesicle signaling in neuronal and immune contexts.



Cancer from cBioPortal

Key Information	Description
Main variant region	Residues 90–200 are highly intolerant to amino acid substitutions, showing functional importance.
Key hotspots	Key sites are residues 45–52 (PSD-95/DLG-binding) and 135–138 (C-mannosylation).
Functional significance	These regions are critical for binding or post-translational modifications in disordered proteins.
Link to core hypothesis	This supports FAM131A's role in regulating vesicle signaling in neuronal and immune cells.

Conclusion:

FAM131A functionally **relevant genomic alterations across multiple cancer types.**

Functional enrichment of co-altered genes highlights **strong associations with vesicle trafficking, synaptic signaling, RNA-binding proteins, ubiquitination machinery, and metabolic regulators.**

These alterations often appear in clusters with pathway nodes directly involved in vesicle maturation and exocytosis, receptor signaling, and redox homeostasis supporting the core hypothesis

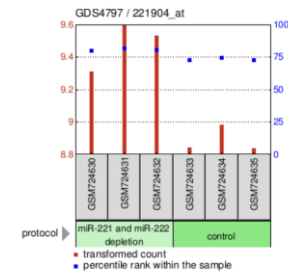
Experiment in GEO

Experiment	Context & Direction	Link to Core Hypothesis
Exp 14: miR-221/222 depletion ↑	Removal of these miRNAs increases FAM131A expression.	Demonstrates post-transcriptional dosage control via miRNA regulation, aligning with the model of fine-tuned adaptor abundance through RNA regulation.
Exp 10: Casiopeina-II-gly ↓ in HeLa cells	Expression decreases under oxidative/metal stress.	Links FAM131A regulation to metabolic and redox cues , consistent with sensing and integrating metabolic states.
Exp 5: CXCL4 ↓ in macrophages	Reduced expression during inflammation.	Shows immune context-dependent regulation , supporting its role in gating vesicle signaling in immune environments.

Summary Sentence

Together, these experiments demonstrate that FAM131A expression is shaped by **miRNA-mediated dosage control**, **metabolic/redox stress response**, and **immune context-dependent regulation**—three pillars of its proposed role as a cytosolic adaptor integrating RNA regulation and metabolism to control vesicle signaling in neuronal and immune systems.

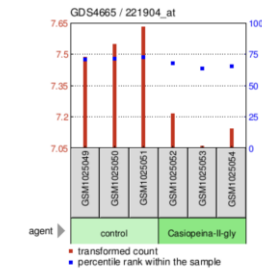
Profile GDS4797 / 221904_at
Title MCF10A mammary epithelial cell line transfected with miR-221/222 mimic
Organism Homo sapiens



[Graph caption help](#)

Sample	Title	Value	Rank
GSM724630	MCF10A_miR_221_222_mimic_1	9.31065	80
GSM724631	MCF10A_miR_221_222_mimic_2	9.59568	82
GSM724632	MCF10A_miR_221_222_mimic_3	9.53392	81
GSM724633	MCF10A_miR_control_1	8.84151	73
GSM724634	MCF10A_miR_control_2	8.98628	75
GSM724635	MCF10A_miR_control_3	8.84023	73

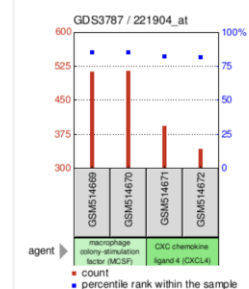
Profile GDS4665 / 221904_at
Title HeLa cell line response to chemotherapeutic Casiopeinas
Organism Homo sapiens



[Graph caption help](#)

Sample	Title	Value	Rank
GSM1025049	HeLa control, biological rep1	7.48838	71
GSM1025050	HeLa control, biological rep2	7.54827	72
GSM1025051	HeLa control, biological rep3	7.63234	73
GSM1025052	HeLa treated with Cas-II-gly, biological rep1	7.21781	68
GSM1025053	HeLa treated with Cas-II-gly, biological rep2	7.05303	64
GSM1025054	HeLa treated with Cas-II-gly, biological rep3	7.14519	66

Profile GDS3787 / 221904_at
Title CXC chemokine ligand 4 effect on monocyte-derived macrophages
Organism Homo sapiens



[Graph caption help](#)

Sample	Title	Value	Rank
GSM514669	MCSF_A	514.1	86
GSM514670	MCSF_B	515.1	86
GSM514671	CXCL4_A	394.9	83
GSM514672	CXCL4_B	343.5	82

Summary

FAM131A:

Role: **cytosolic adaptor**

Mechanism: **RNA regulation, ubiquitin signaling, and metabolic cues**

Function: **vesicle-dependent signaling** in both neuronal and immune systems.

Evidence:

from protein interactions, motif architecture, expression profiles, cancer co-alterations, and experimental data shows it is **a flexible, motif-rich scaffold positioned at key pathway intersections**. Its regulation is shaped by metabolic and redox stress responses, and immune context.

Future experiment

1. Validate key motifs – Confirm the roles of the **phospho-dependent MAGUK GK-like motif** and **C-mannosylation sequence** in vesicle maturation and receptor signaling.

2. Study miRNA regulation – Determine how **miRNA dosage control** influences FAM131A expression and downstream vesicle signaling.

3. Link to metabolic/redox stress – Test how **metabolic cues** and **ROS changes** modulate localization, partner binding, and vesicle transport.

Experiment design

Does FAM131A boost vesicle secretion?

Idea: Overexpress FAM131A in a friendly cell line (HEK293T). Measure bulk secretory/exocytosis using a **secreted luciferase** reporter (Gaussia). If FAM131A helps vesicle-dependent trafficking, you'll see **more reporter in the medium**.

Cell & plasmids

Cells: HEK293T

Plasmids:

- pCMV-FAM131A

- Secreted reporter (e.g., Gaussia luciferase under CMV).

- Empty vector (control).

Transfect

- Group 1: EV + Gaussia (baseline control)

- Group 2: FAM131A + Gaussia (test)

Collect media & cells

- At 24 h, 36 h, 48 h: collect **media** (for secreted luciferase).

- Lyse cells (for **reporter mRNA** and **reporter intracellular activity**) to control for expression differences.

Readouts

- Gaussia activity in media** (luminometer) → secretion output.

- Gaussia in lysate** and **RT-qPCR for reporter mRNA** → ensure secretion change isn't just expression.

- Western/RT-qPCR** to confirm **FAM131A overexpression**.

Expected result

- FAM131A OE > EV**: higher extracellular/total reporter ratio (↑ secretion).

- Motif mutant**: reduced or no increase vs WT → supports “scaffold/vesicle” mechanism.

THANK YOU